STOR 455 Group Project (Due 11:59pm on April 5th)

Luigi's Linear Models

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The Prediction (Required)

Our prediction of the cumulative domestic box office of "Super Mario Galaxy" by May 1, 2026 is $485.3 million.

Summary of Justification (Required)

Summary: We built a linear regression model using 27 animated sequels, predicting log domestic total from log previous film earnings and log budget (Adj. $R^2 = 0.469$). Applying a 0.80 multiplier for 30-day earnings, our model predicts Super Mario Galaxy will earn approximately $485 million domestically in its first 30 days. This prediction is based on the strong positive relationships observed between the previous film's performance, budget, and domestic earnings in our sample of animated sequels. The model's predictors are statistically significant, and the assumptions of linear regression are reasonably satisfied.

Analysis

```r
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3

## Warning: package 'tibble' was built under R version 4.4.3

## Warning: package 'tidyr' was built under R version 4.4.3

## Warning: package 'readr' was built under R version 4.4.3

## Warning: package 'purrr' was built under R version 4.4.3

## Warning: package 'dplyr' was built under R version 4.4.3

## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
library(stringr)
library(DataExplorer)
```

```
## Warning: package 'DataExplorer' was built under R version 4.4.3
```

```r
library(funModeling)
```

```
## Warning: package 'funModeling' was built under R version 4.4.3

## Loading required package: Hmisc

## Warning: package 'Hmisc' was built under R version 4.4.3

##
## Attaching package: 'Hmisc'
##
## The following objects are masked from 'package:dplyr':
##
##     src, summarize
##
## The following objects are masked from 'package:base':
##
##     format.pval, units
##
## funModeling v.1.9.6 :)
## Examples and tutorials at livebook.datascienceheroes.com
##  / Now in Spanish: librovivodecienciadedatos.ai
```

```r
movies <- read.csv('data/animated_movies.csv') |>
  clean_names()
```

```r
glimpse(movies)
```

```
## Rows: 27
## Columns: 9
## $ movie              <chr> "Inside Out 2", "Incredibles 2", "Finding Dory", "F~
## $ domestic_total     <int> 652980194, 608581744, 486295561, 477373578, 4604052~
## $ prev_film_domestic <int> 356461711, 261441092, 339714978, 400953009, 2487570~
## $ year               <int> 2024, 2018, 2016, 2019, 2024, 2019, 2010, 2025, 201~
## $ budget             <int> 200000000, 200000000, 200000000, 150000000, 1500000~
## $ is_video_game      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ is_spring_release  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, ~
## $ rotten_tomatoes    <int> 98, 97, 99, 89, 95, 98, 100, 98, 80, 75, 58, 55, 96~
## $ pocorn_meter       <int> 89, 75, 86, 85, 89, 90, 87, 92, 83, 85, 53, 49, 90,~
```

```r
movies <- movies |>
  mutate(across(c(domestic_total, prev_film_domestic, budget, year), as.numeric)) |>
  mutate(across(c(is_video_game, is_spring_release), as.factor))

movies$log_domestic <- log(movies$domestic_total)
movies$log_prev <- log(movies$prev_film_domestic)
movies$log_budget <- log(movies$budget)

glimpse(movies)
```

```
## Rows: 27
## Columns: 12
## $ movie              <chr> "Inside Out 2", "Incredibles 2", "Finding Dory", "F~
## $ domestic_total     <dbl> 652980194, 608581744, 486295561, 477373578, 4604052~
## $ prev_film_domestic <dbl> 356461711, 261441092, 339714978, 400953009, 2487570~
## $ year               <dbl> 2024, 2018, 2016, 2019, 2024, 2019, 2010, 2025, 201~
## $ budget             <dbl> 2.00e+08, 2.00e+08, 2.00e+08, 1.50e+08, 1.50e+08, 2~
## $ is_video_game      <fct> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ is_spring_release  <fct> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, ~
## $ rotten_tomatoes    <int> 98, 97, 99, 89, 95, 98, 100, 98, 80, 75, 58, 55, 96~
## $ pocorn_meter       <int> 89, 75, 86, 85, 89, 90, 87, 92, 83, 85, 53, 49, 90,~
## $ log_domestic       <dbl> 20.29706, 20.22664, 20.00233, 19.98381, 19.94762, 1~
## $ log_prev           <dbl> 19.69174, 19.38172, 19.64362, 19.80935, 19.33199, 1~
## $ log_budget         <dbl> 19.11383, 19.11383, 19.11383, 18.82615, 18.82615, 1~
```

#Linear Regression Analysis: Best subsets:

```r
library(leaps)
best_subset <- regsubsets(log_domestic ~ log_prev + year + is_video_game + is_spring_release + log_budge
sub_summary <- summary(best_subset)
sub_summary
```

```
## Subset selection object
## Call: regsubsets.formula(log_domestic ~ log_prev + year + is_video_game +
##     is_spring_release + log_budget + rotten_tomatoes + pocorn_meter,
##     data = movies, nvmax = 7)
## 7 Variables  (and intercept)
```

```
##                     Forced in Forced out
## log_prev                FALSE      FALSE
## year                    FALSE      FALSE
## is_video_game1          FALSE      FALSE
## is_spring_release1      FALSE      FALSE
## log_budget              FALSE      FALSE
## rotten_tomatoes         FALSE      FALSE
## pocorn_meter            FALSE      FALSE
## 1 subsets of each size up to 7
## Selection Algorithm: exhaustive
##          log_prev year is_video_game1 is_spring_release1 log_budget
## 1  ( 1 ) "*"      " "  " "            " "                " "
## 2  ( 1 ) "*"      " "  " "            " "                "*"
## 3  ( 1 ) "*"      "*"  " "            " "                "*"
## 4  ( 1 ) "*"      "*"  " "            " "                "*"
## 5  ( 1 ) "*"      "*"  " "            "*"                "*"
## 6  ( 1 ) "*"      "*"  " "            "*"                "*"
## 7  ( 1 ) "*"      "*"  "*"            "*"                "*"
##          rotten_tomatoes pocorn_meter
## 1  ( 1 ) " "             " "
## 2  ( 1 ) " "             " "
## 3  ( 1 ) " "             " "
## 4  ( 1 ) "*"             " "
## 5  ( 1 ) "*"             " "
## 6  ( 1 ) "*"             "*"
## 7  ( 1 ) "*"             "*"
```

```r
data.frame(
  Predictors = 1:7,
  AdjR2      = round(sub_summary$adjr2, 4),
  Cp         = round(sub_summary$cp, 4),
  BIC        = round(sub_summary$bic, 4)
)
```

```
##   Predictors  AdjR2     Cp     BIC
## 1          1 0.3688 3.9123 -6.8914
## 2          2 0.4685 0.7563 -9.3380
## 3          3 0.4874 1.1095 -8.1673
## 4          4 0.4793 2.5357 -5.6532
## 5          5 0.4645 4.1781 -2.8562
## 6          6 0.4428 6.0071  0.1980
## 7          7 0.4137 8.0000  3.4837
```

Interpretation: It appears that a model with 2 predictors is the most optimal. Has lowest Cp and BIC, second highest R^2. The jump from predictors 2 to 3 is very small for R^2.

##Change this

```r
# Quadratics
model_q1 <- lm(log_domestic ~ log_prev + log_budget + I(log_prev^2), data = movies)
model_q2 <- lm(log_domestic ~ log_prev + log_budget + I(log_budget^2), data = movies)

# Interaction between the two winning predictors
model_int <- lm(log_domestic ~ log_prev * log_budget, data = movies)
```

```r
# Base Model
model_base <- lm(log_domestic ~ log_prev + log_budget, data = movies)

anova(model_base, model_q1)   # does logPrev² help?
```

```
## Analysis of Variance Table
##
## Model 1: log_domestic ~ log_prev + log_budget
## Model 2: log_domestic ~ log_prev + log_budget + I(log_prev^2)
##   Res.Df    RSS Df Sum of Sq     F Pr(>F)
## 1     24 2.9136
## 2     23 2.9102  1 0.0034185 0.027 0.8709
```

```r
anova(model_base, model_q2)   # does logBudget² help?
```

```
## Analysis of Variance Table
##
## Model 1: log_domestic ~ log_prev + log_budget
## Model 2: log_domestic ~ log_prev + log_budget + I(log_budget^2)
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     24 2.9136
## 2     23 2.8476  1  0.065973 0.5329 0.4728
```

```r
anova(model_base, model_int)  # does the interaction help?
```

```
## Analysis of Variance Table
##
## Model 1: log_domestic ~ log_prev + log_budget
## Model 2: log_domestic ~ log_prev * log_budget
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1     24 2.9136
## 2     23 2.4961  1   0.41755 3.8475 0.06203 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: None of the quadratic terms nor the interaction term significantly improved the model (all $p > 0.05$), confirming that the base model with log_prev and log_budget is the most appropriate.

```r
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##     recode

## The following object is masked from 'package:purrr':
##
##     some
```

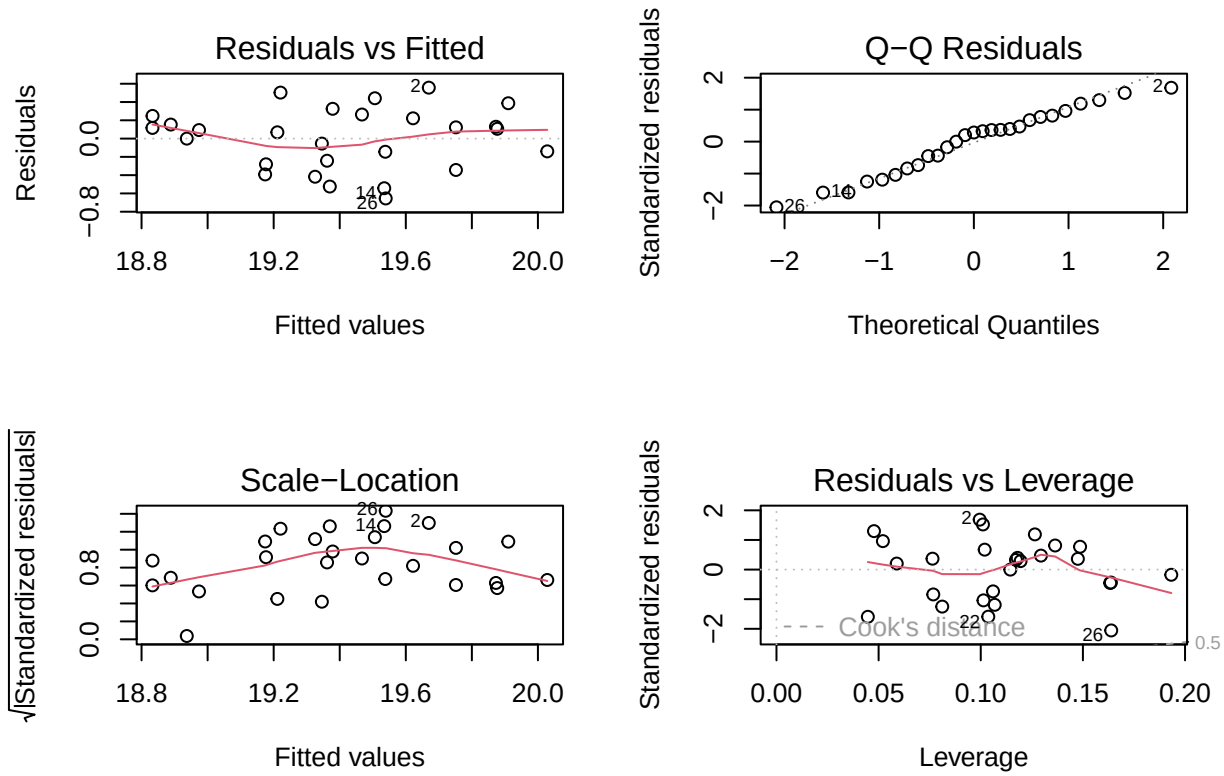
```r
summary(model_base)
```

```
##
## Call:
## lm(formula = log_domestic ~ log_prev + log_budget, data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.65484 -0.26174  0.09319  0.23495  0.55702
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -3.6594     4.6430  -0.788 0.438322
## log_prev      0.7762     0.2035   3.814 0.000841 ***
## log_budget    0.4334     0.1817   2.385 0.025332 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3484 on 24 degrees of freedom
## Multiple R-squared:  0.5094, Adjusted R-squared:  0.4685
## F-statistic: 12.46 on 2 and 24 DF,  p-value: 0.0001946
```

```r
vif(model_base)
```

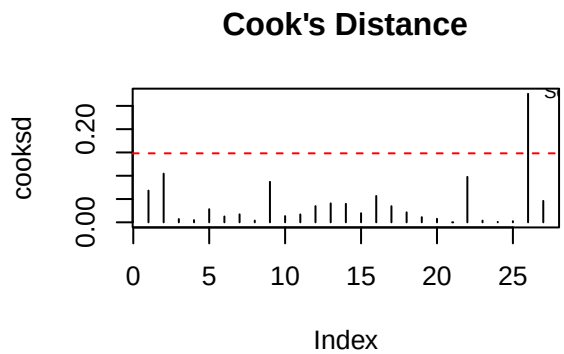
```
##   log_prev log_budget
##    1.04235    1.04235
```

```r
par(mfrow = c(2, 2))
plot(model_base)
```




```r
cooksd <- cooks.distance(model_base)
plot(cooksd, type = "h", main = "Cook's Distance")
abline(h = 4/nrow(movies), col = "red", lty = 2)
text(which(cooksd > 4/nrow(movies)),
     cooksd[cooksd > 4/nrow(movies)],
     labels = movies$movie[cooksd > 4/nrow(movies)], cex = 0.7, pos = 4)
shapiro.test(residuals(model_base))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(model_base)
## W = 0.96543, p-value = 0.4867
```



Interpretation of Findings: The final model, log_domestic ~ log_prev + log_budget, is statistically significant overall ($F = 12.46$, $p < 0.001$). Both predictors are significant: log_prev ($p < 0.001$) and log_budget ($p = 0.025$). The model explains about 46.9% of the variability in log domestic total (Adjusted $R^2 = 0.4685$). For every 1-unit increase in log of the previous film's domestic earnings, log domestic total increases by 0.776, holding budget constant. Similarly, a 1-unit increase in log budget is associated with a 0.433 increase in log domestic total, holding the previous film's performance constant. Both VIF values are approximately 1.04, which is extremely close to 1, indicating no meaningful multicollinearity between log_prev and log_budget. The predictors are essentially independent of each other.

Interpretation of Graphs: The residuals appear roughly randomly scattered around zero with no strong systematic pattern, suggesting the linearity assumption is reasonably satisfied. The red smoothing line is relatively flat. Observations 14 and 26 appear as potential outliers with large negative residuals and warrant further investigation.The standardized residuals follow the diagonal reference line closely across most of the range, supporting the normality assumption. Observations 26 and 14 deviate slightly at the lower tail, but the departure is not severe.

Shapiro-Wilk Test: $W = 0.965$, $p = 0.487$. Since $p > 0.05$, we fail to reject the null hypothesis of normality. Combined with the Q-Q plot, there is no meaningful evidence that the normality assumption is violated. Note that with only $n = 27$ observations, this test has limited power, so the visual QQ plot provides additional reassurance.

```
mario_new <- data.frame(
  log_prev   = log(574930000),
  log_budget = log(175000000)
)
pred <- predict(model_base, newdata = mario_new,
                interval = "prediction", level = 0.95)
```

```r
full_run <- exp(pred)
day30_prediction <- full_run * 0.80
day30_prediction
```

```
##         fit       lwr        upr
## 1 485297425 215121283 1094794466
```

Using the final model with the previous film's domestic earnings of $574,930,000 (The Super Mario Bros. Movie) and a production budget of $175,000,000, the model predicts a 30-day domestic total of approximately $485.3 million (95% prediction interval: $215.1 million to $1.095 billion). This was obtained by generating a full-run prediction on the log scale, back-transforming via exponentiation, and applying a 0.80 multiplier to approximate the 30-day earnings, consistent with the typical pattern for animated sequels where roughly 78–82% of total domestic revenue is earned within the first 30 days. The wide prediction interval reflects the inherent variability in animated sequel box office performance and the relatively small sample size, and should be interpreted with caution.